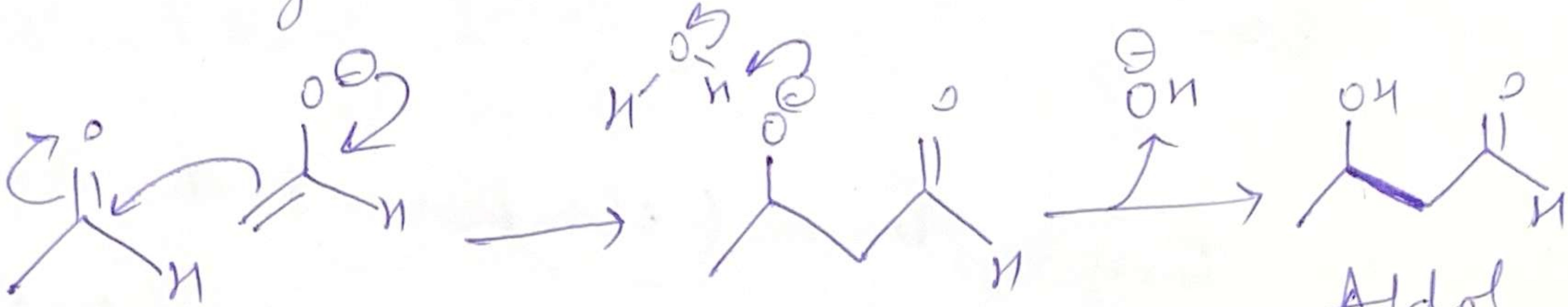
(1) Aldol Reaction

Acetaldehyde



Aldol

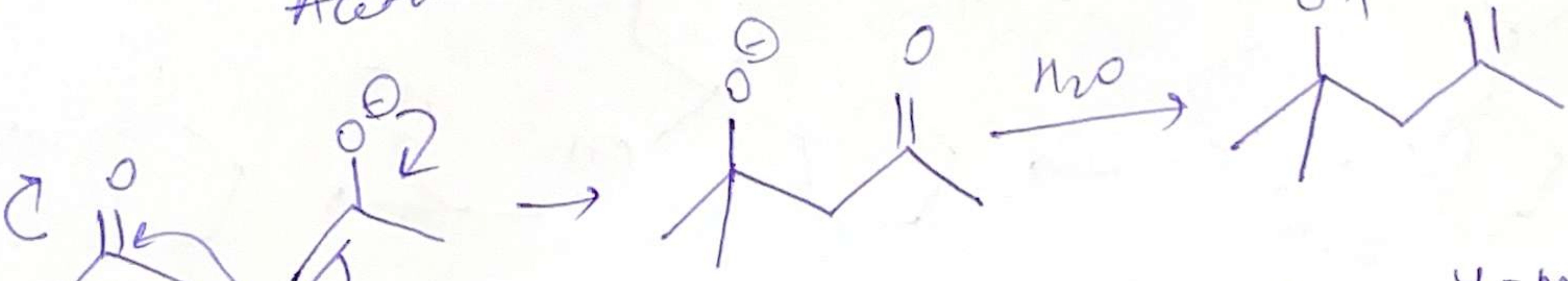
3-hydroxy Butanal ~~butanol~~

Base is regenerated in last step.

Therefore, it's a catalyst.

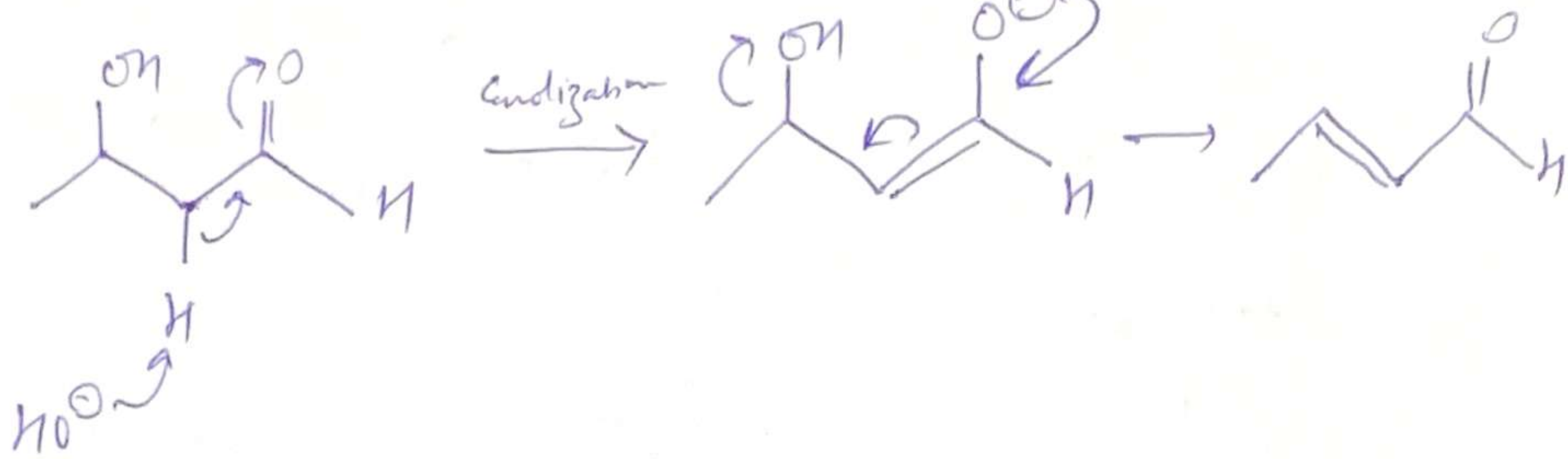
(2) Occurs with ketones as well.

Acetone



4-hydroxy-4-methyl pentan-2-one

If base is present in excess →

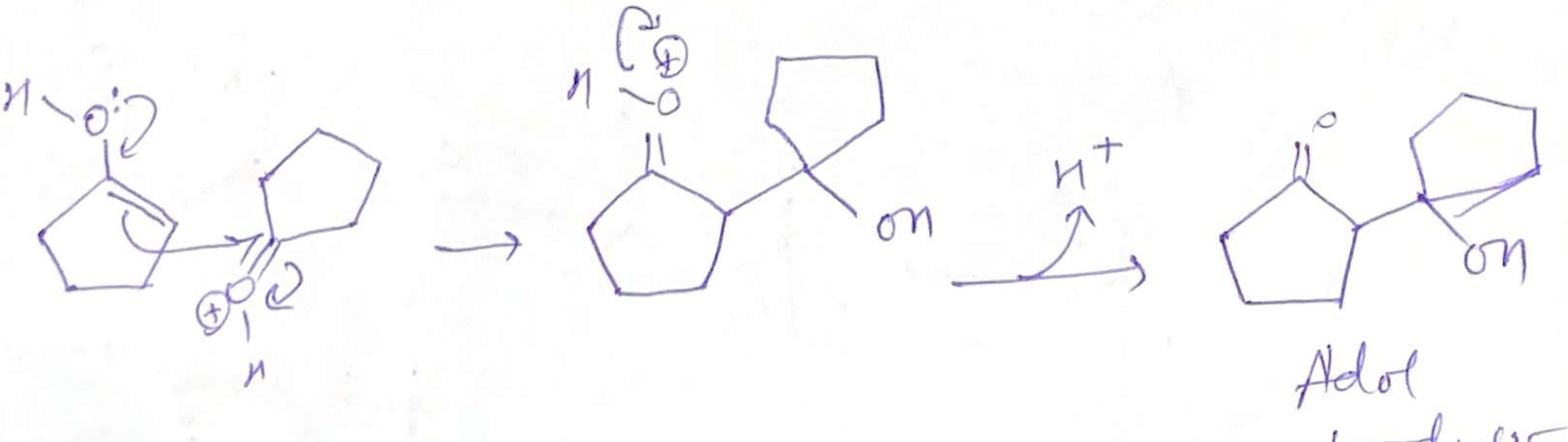


Base → Excess (elimination product)

Structure → some compd. are easy to stop at the aldol stage

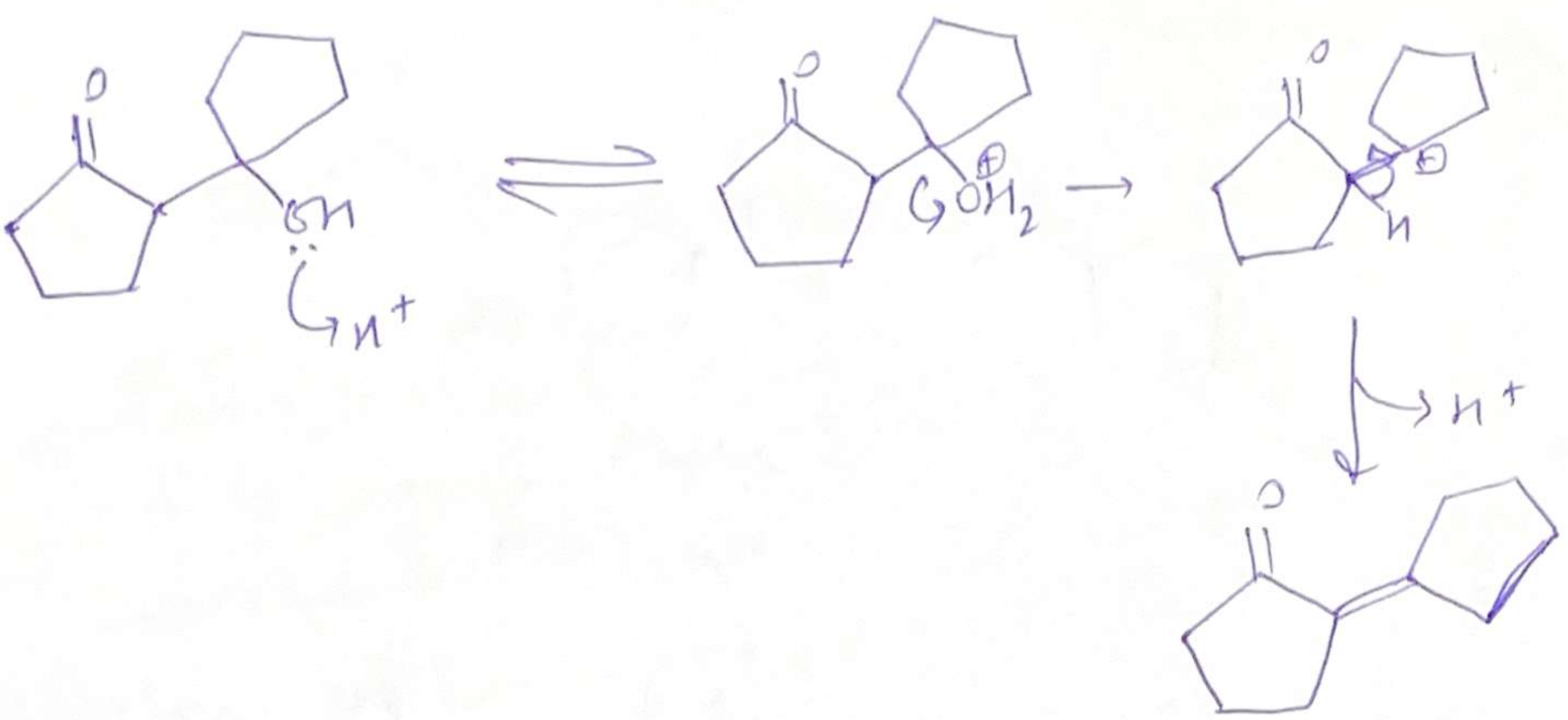
Acid-catalyzed aldol reacn commonly give unsaturated products instead of aldols

enol
(less nu⊖ than enolates)



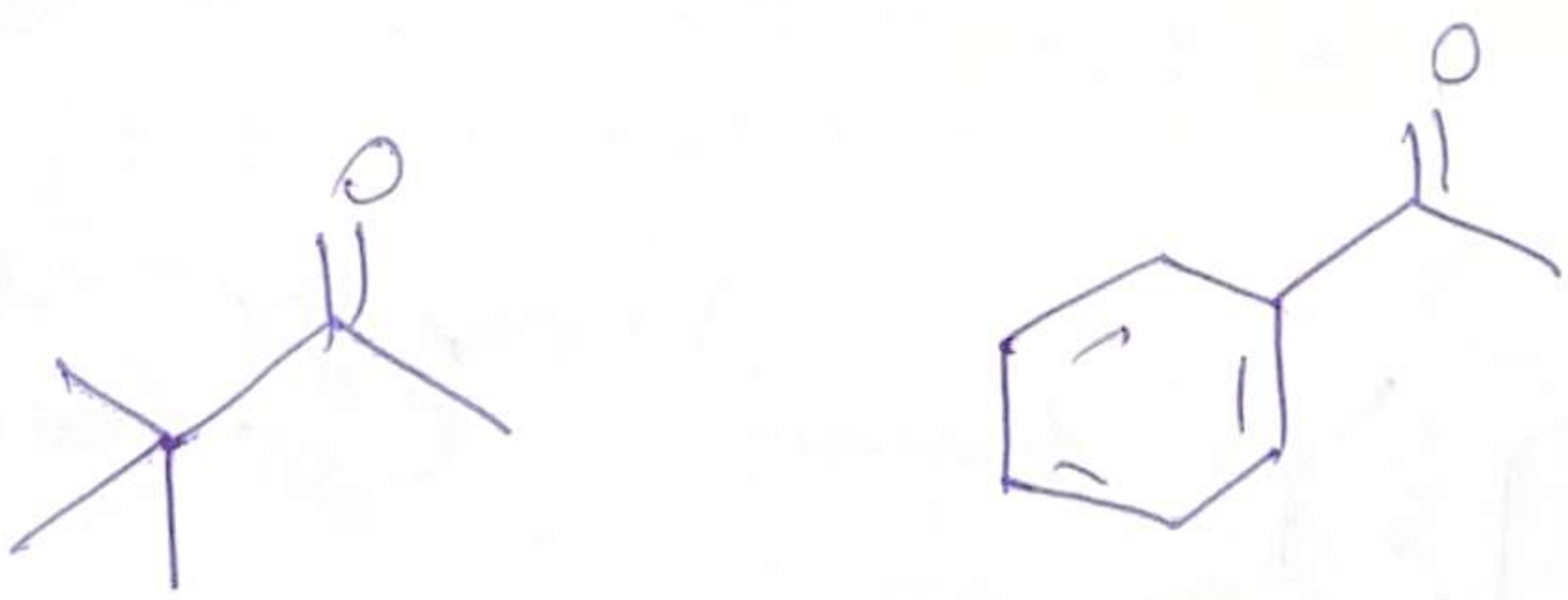
Adol products

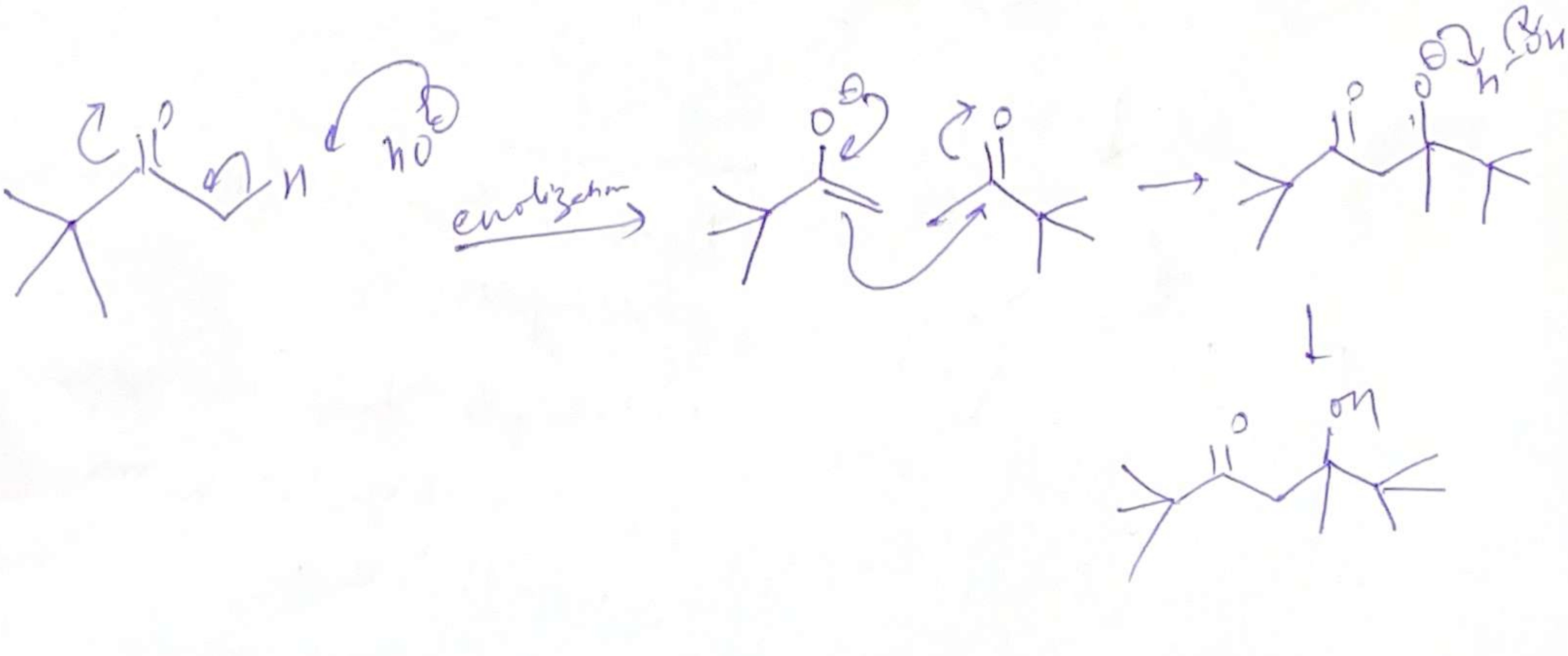
(t-alcohol)



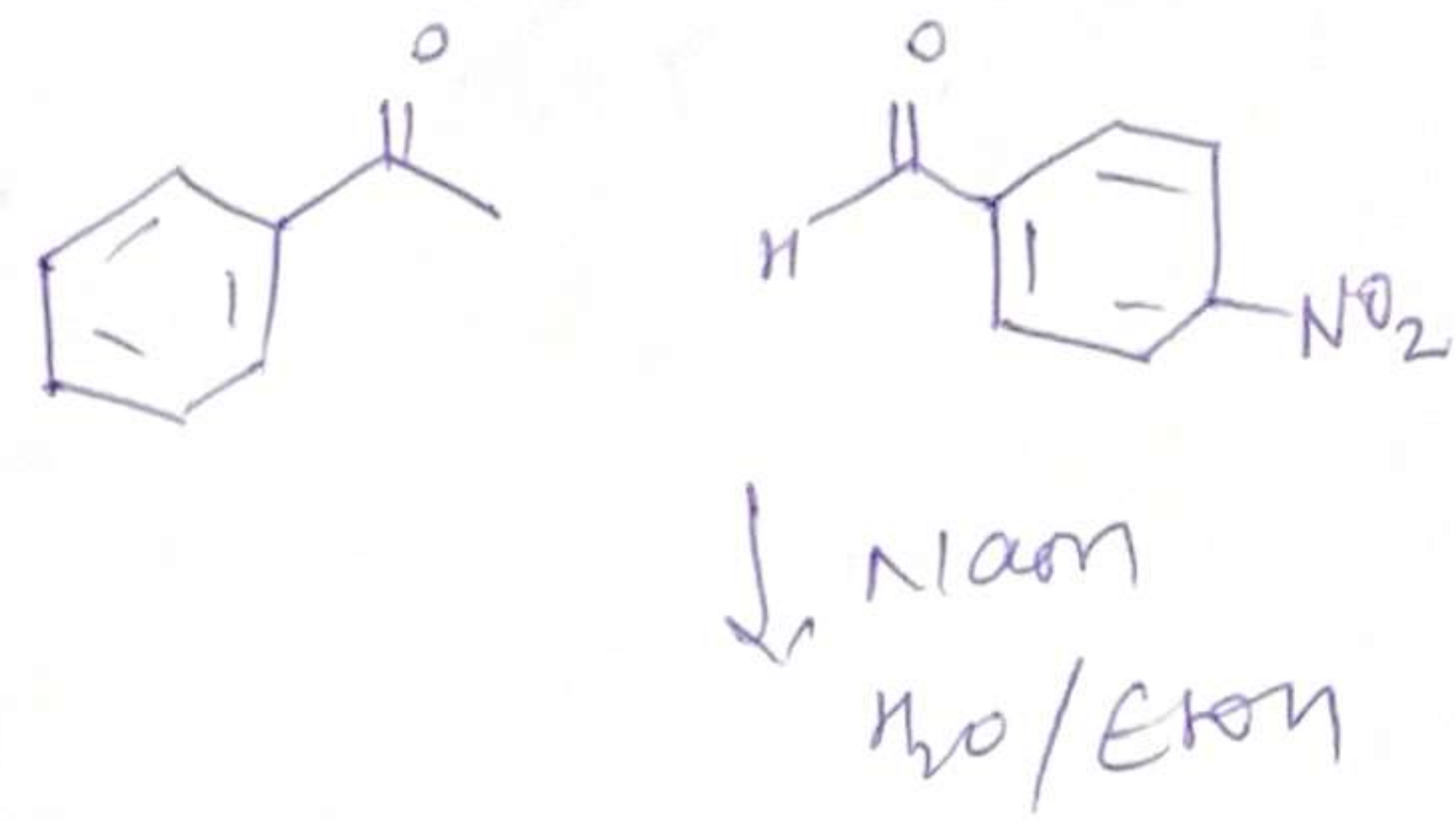
Aldol Condensation → Loss of small molecule usually water

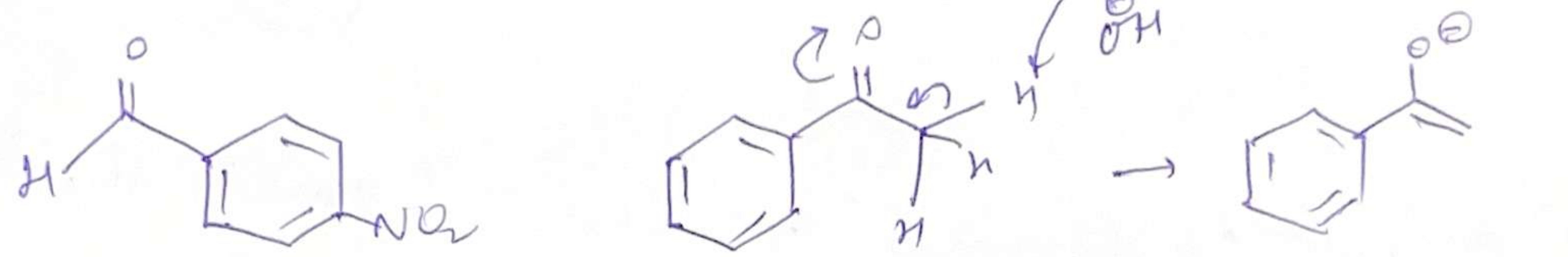
Aldol reac^n of unsymmetrical ketones



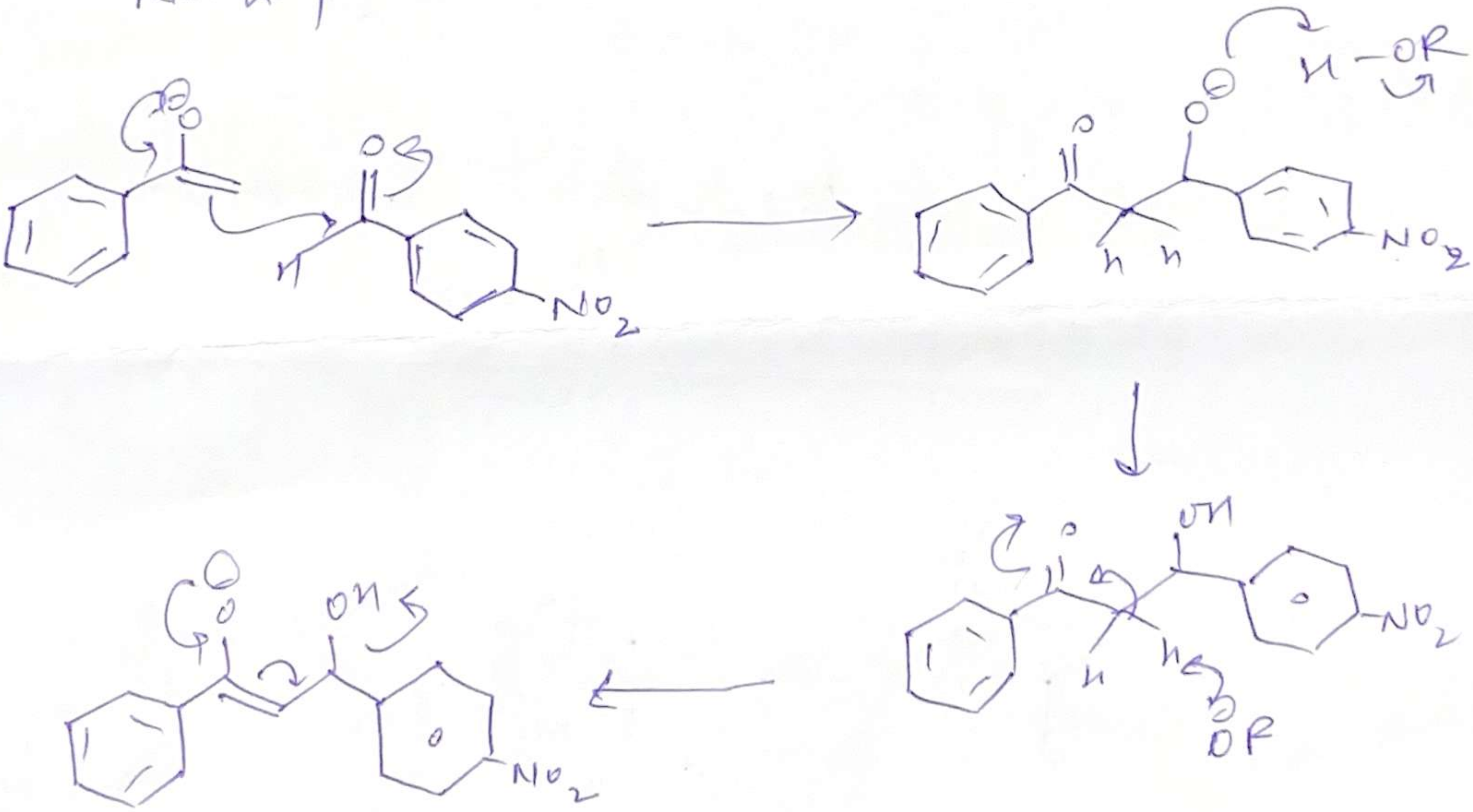


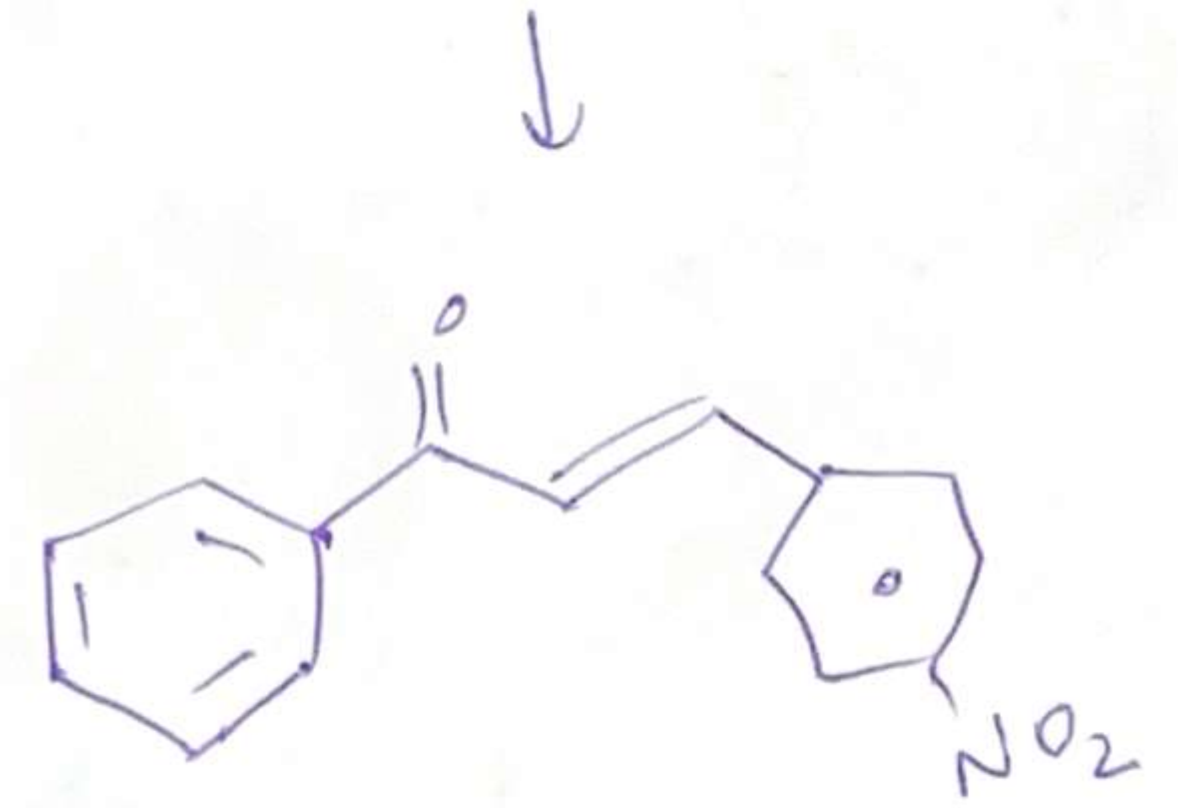
Cross-Condensation:





No. α-protons

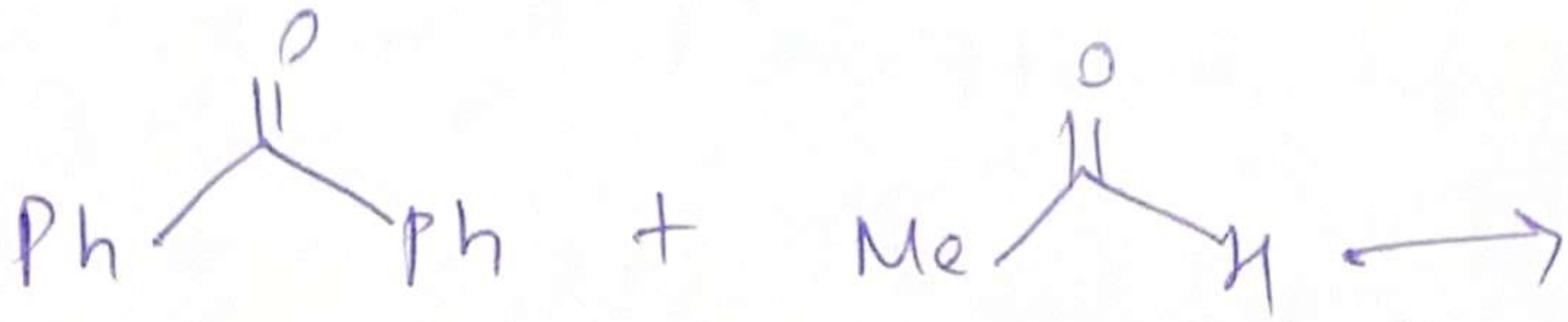


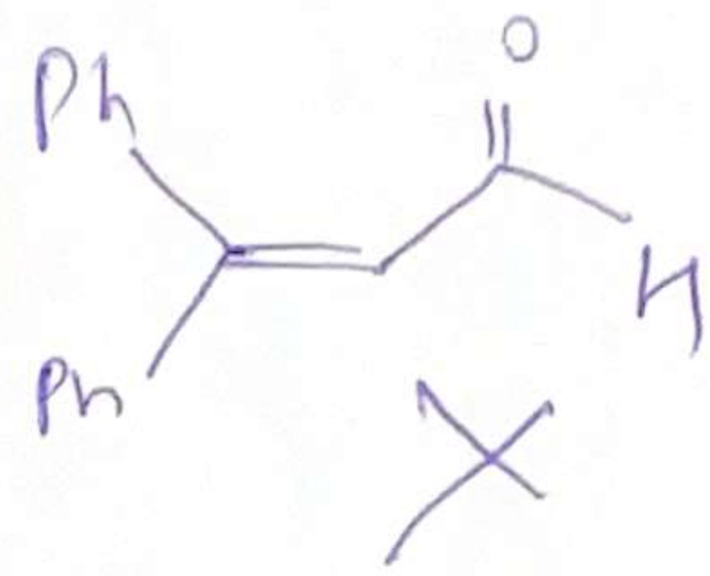


Ketones less reactive than aldehydes

Aldehyde has an ~~electro~~ EWG making it more electrophilic.

Enolate selects the better electrophile i.e. aldehyde





→ Ketae is very hindered & cayingated.
→ Less electrophilic
→ Therefore enolate will chose the best electrophile & attack aldehyde.

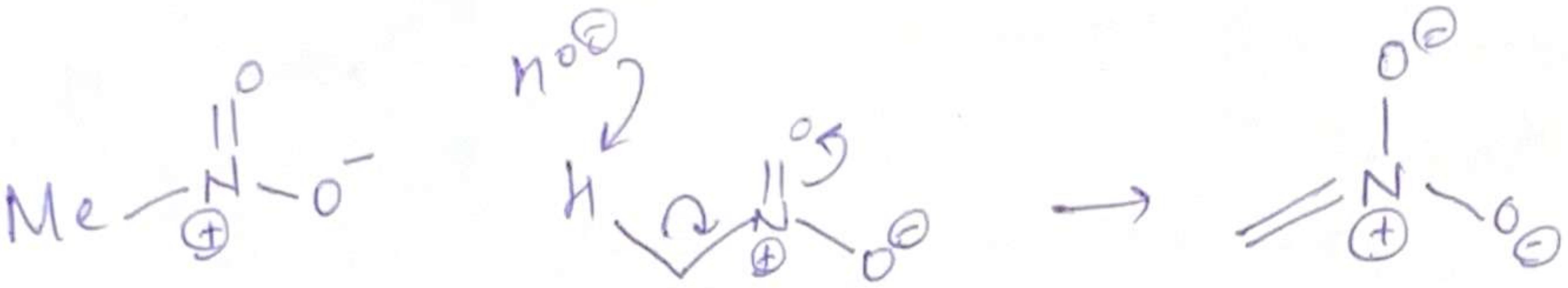
For Successful crossed aldol reac^n

① only one partner must be Capable of enolization.

② The other partner must be incapable of enolization & be more electrophilic than the enolizable partner.

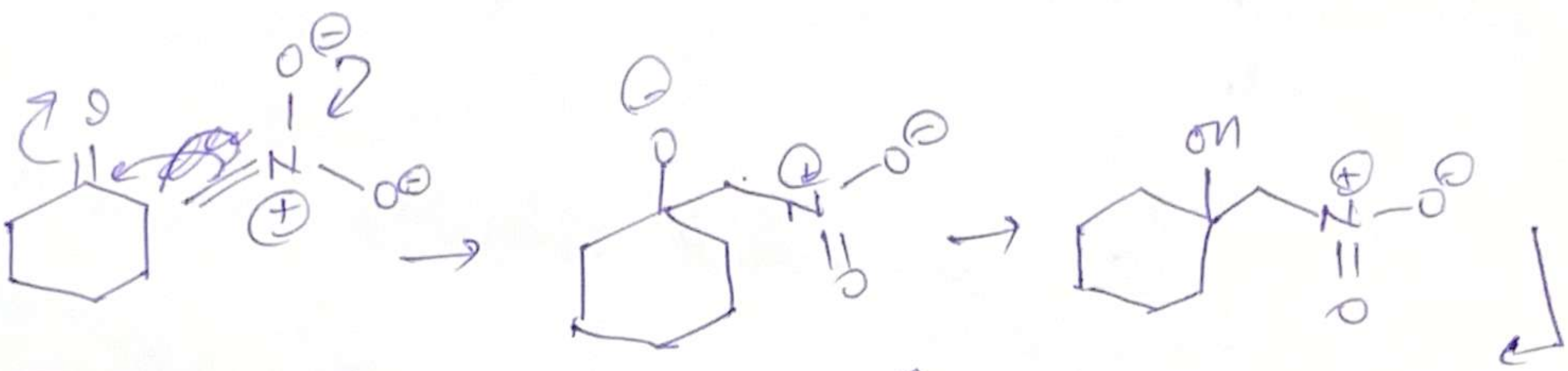
Compounds that can enolize but are not electrophilic

Henry's Reaction



enolization

Nitronate



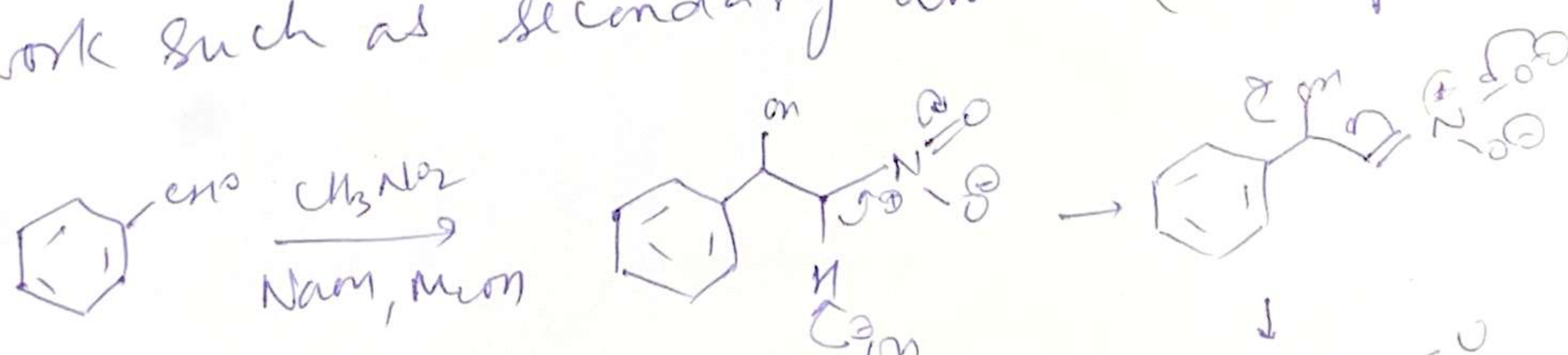
self aldol of ketone doesn't happen because base prefers to remove proton from nitromethane

pKa of ketone = 20

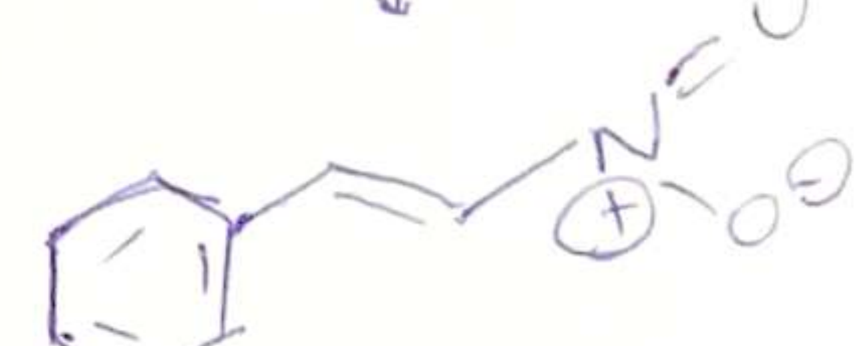
Nitromethane = 10

NaOH (pKaH = 15.7)

Not necessary to use NaOH. Even amines can work such as secondary amines (mostly used)



Difficult to avoid Elimination step specially with Aryl Carbonyls.

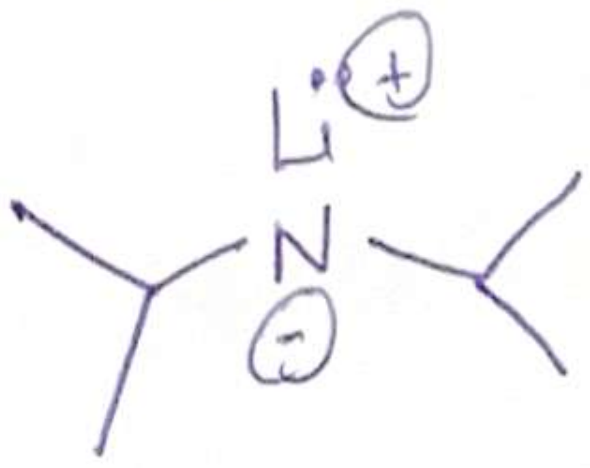


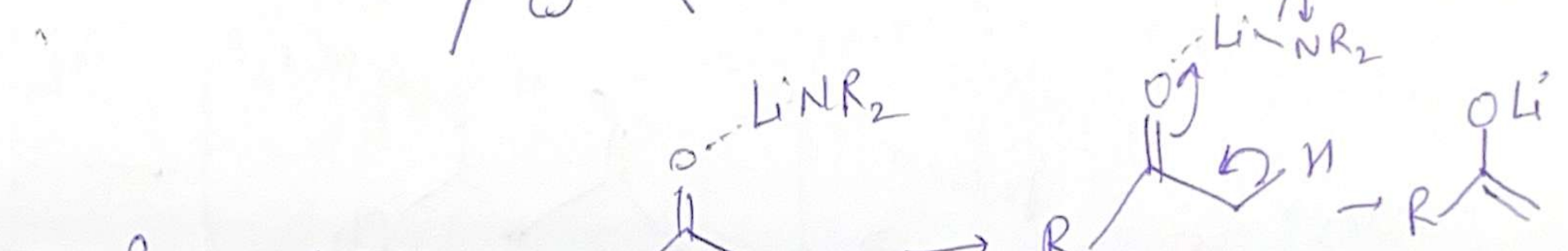
Cross-aldol

- only one enolizable component
- only one set of enolizable protons
- A carbonyl electrophile more reactive than the compound being enolized.

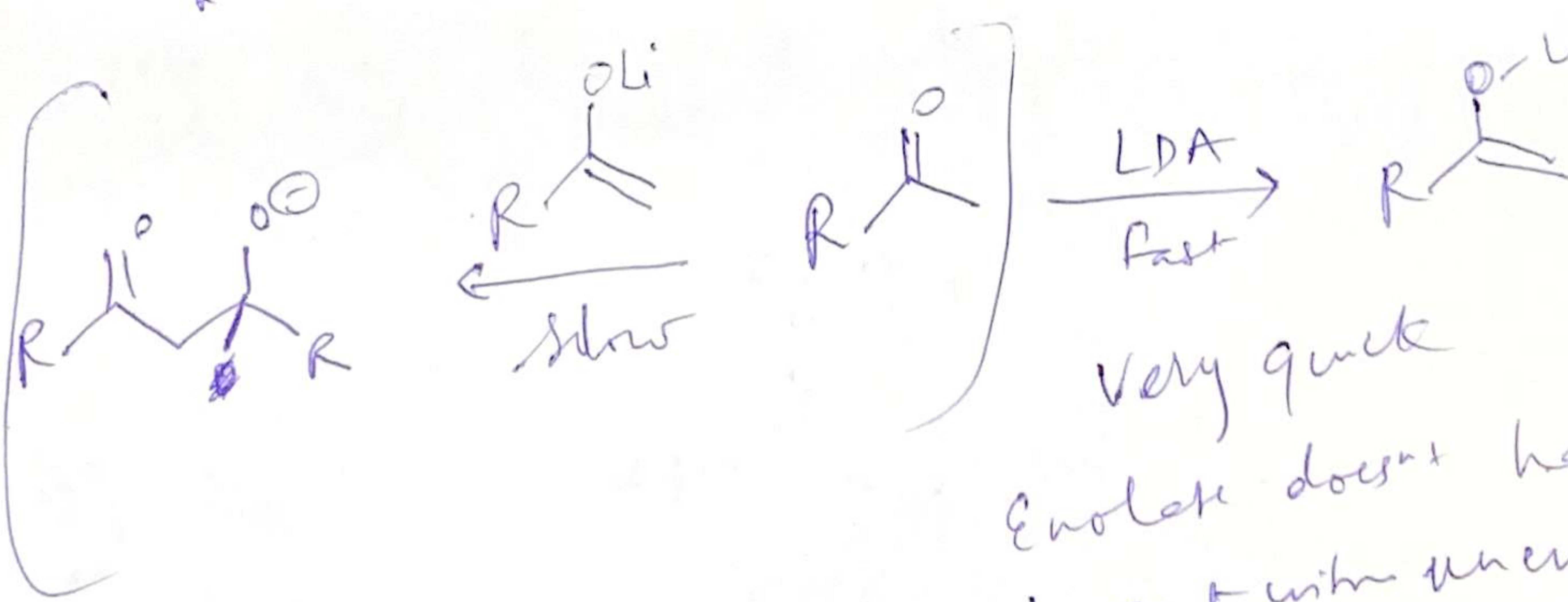
In case which fail to meet these criteria, specific enol equivalents must be used.

(1) Lithium enolates:

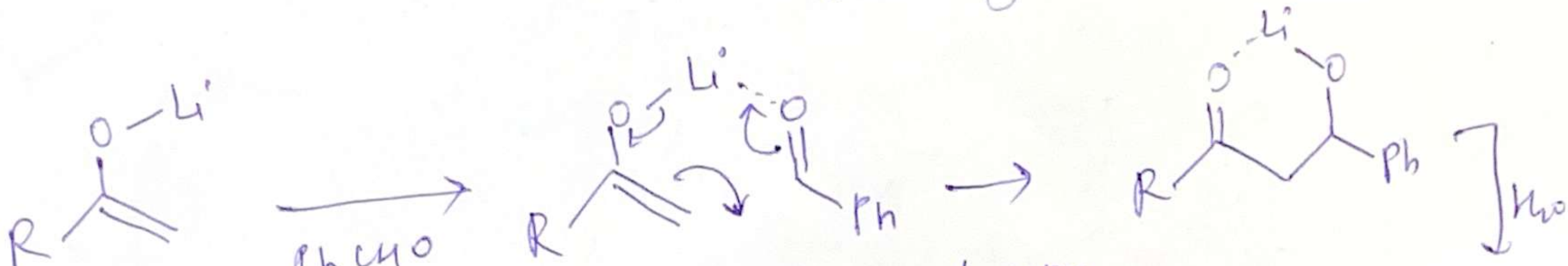




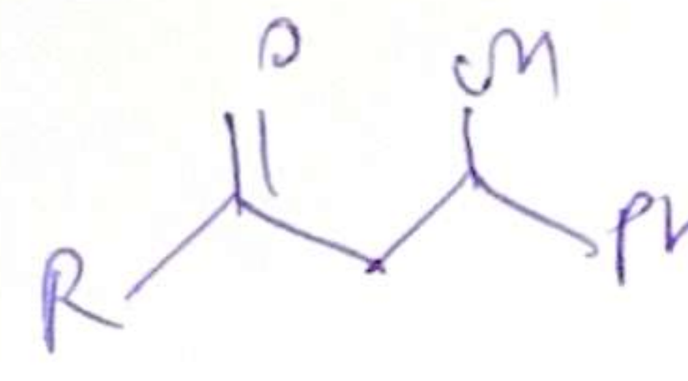
Lithium enolate



Very quick
Enolate doesn't have chance to react with unenolized carbonyl compound.

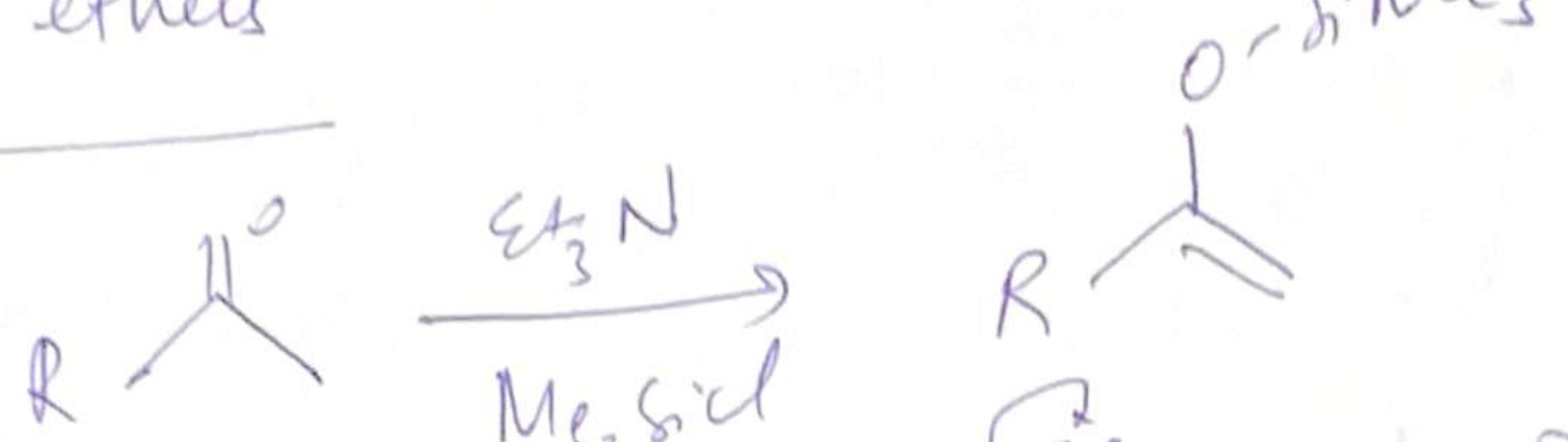


Cyclic Mechanism

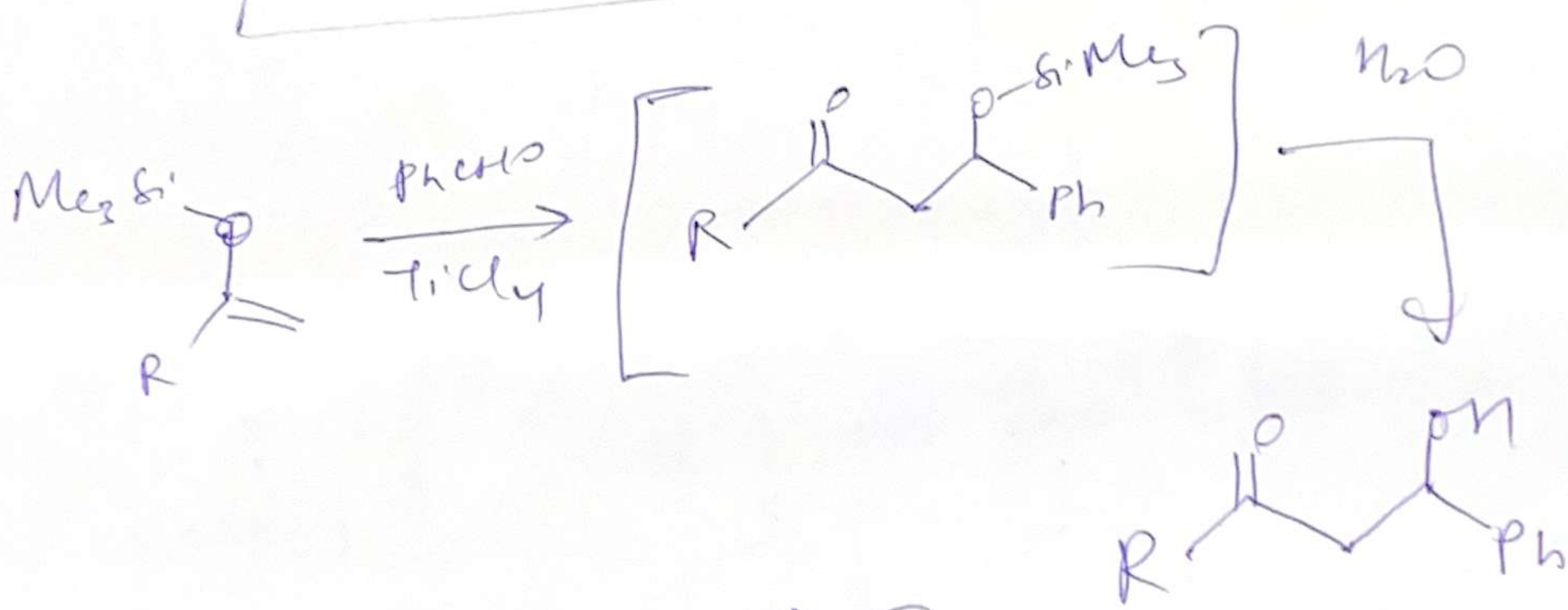


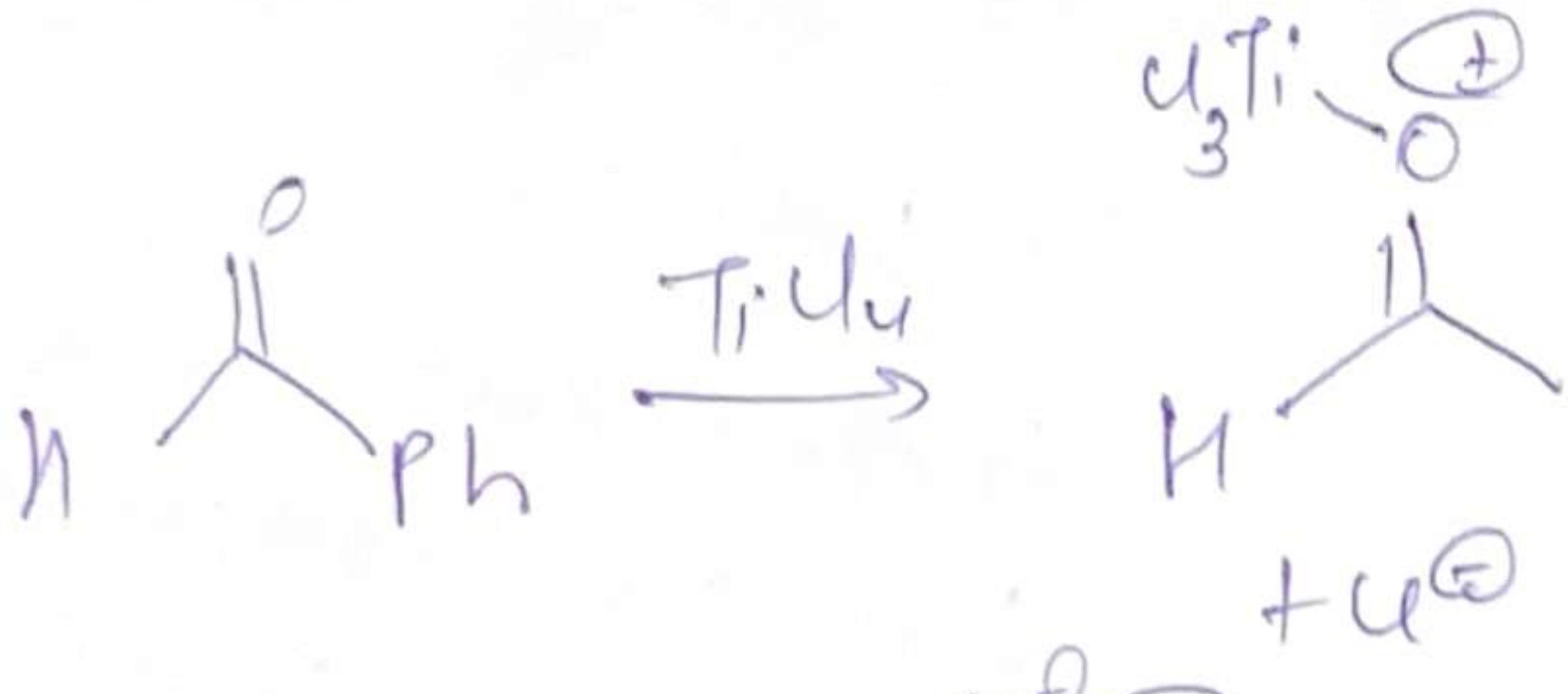
Works well even if the electrophilic partner is enolizable

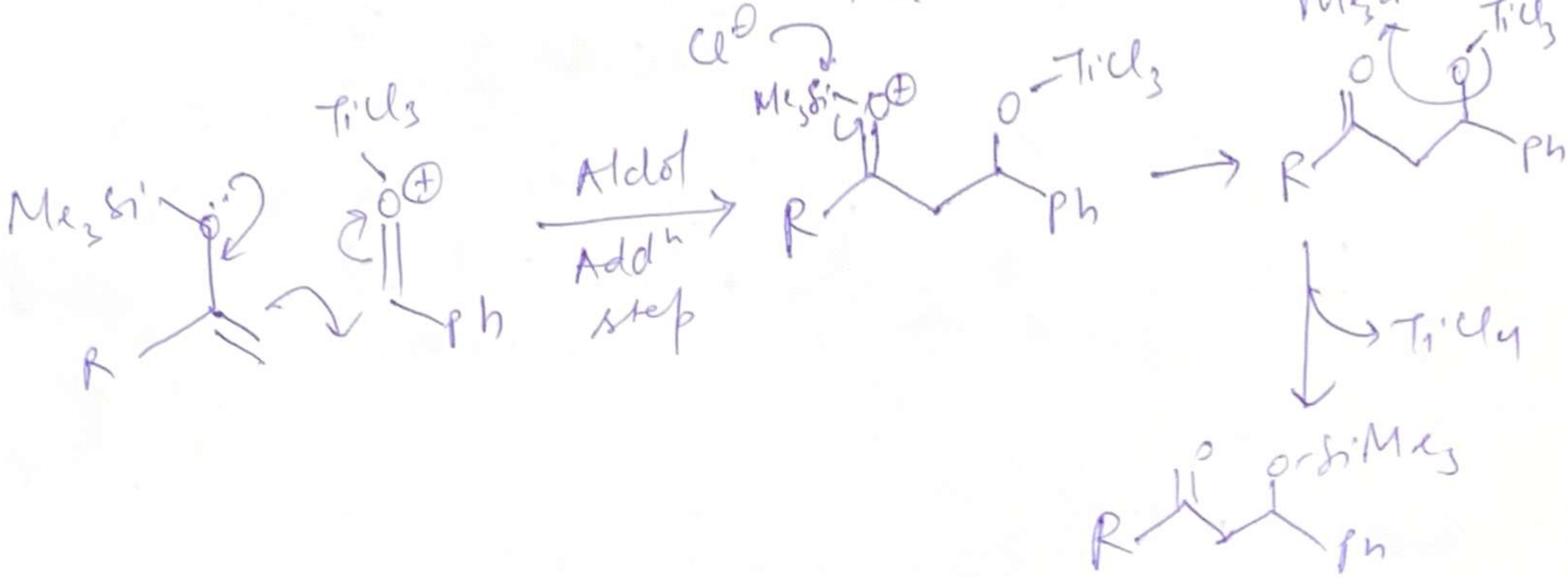
Silyl enol ethers



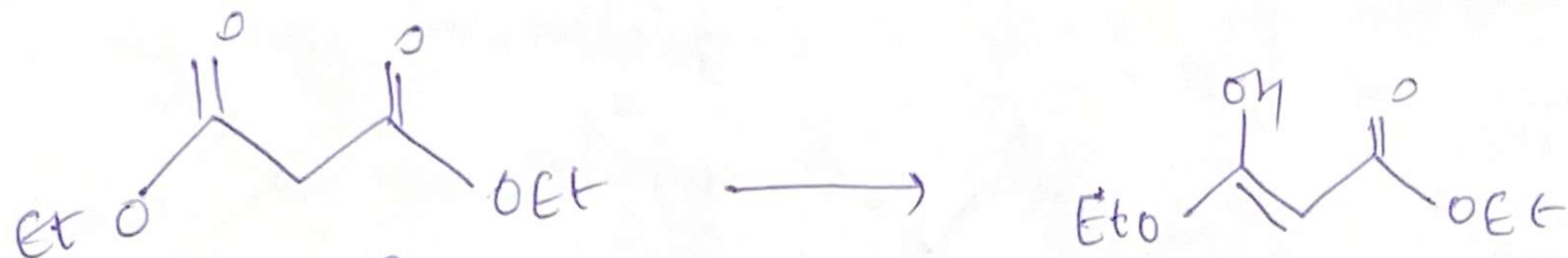
$\boxed{\text{Less reactive than Lithium enolates}}$







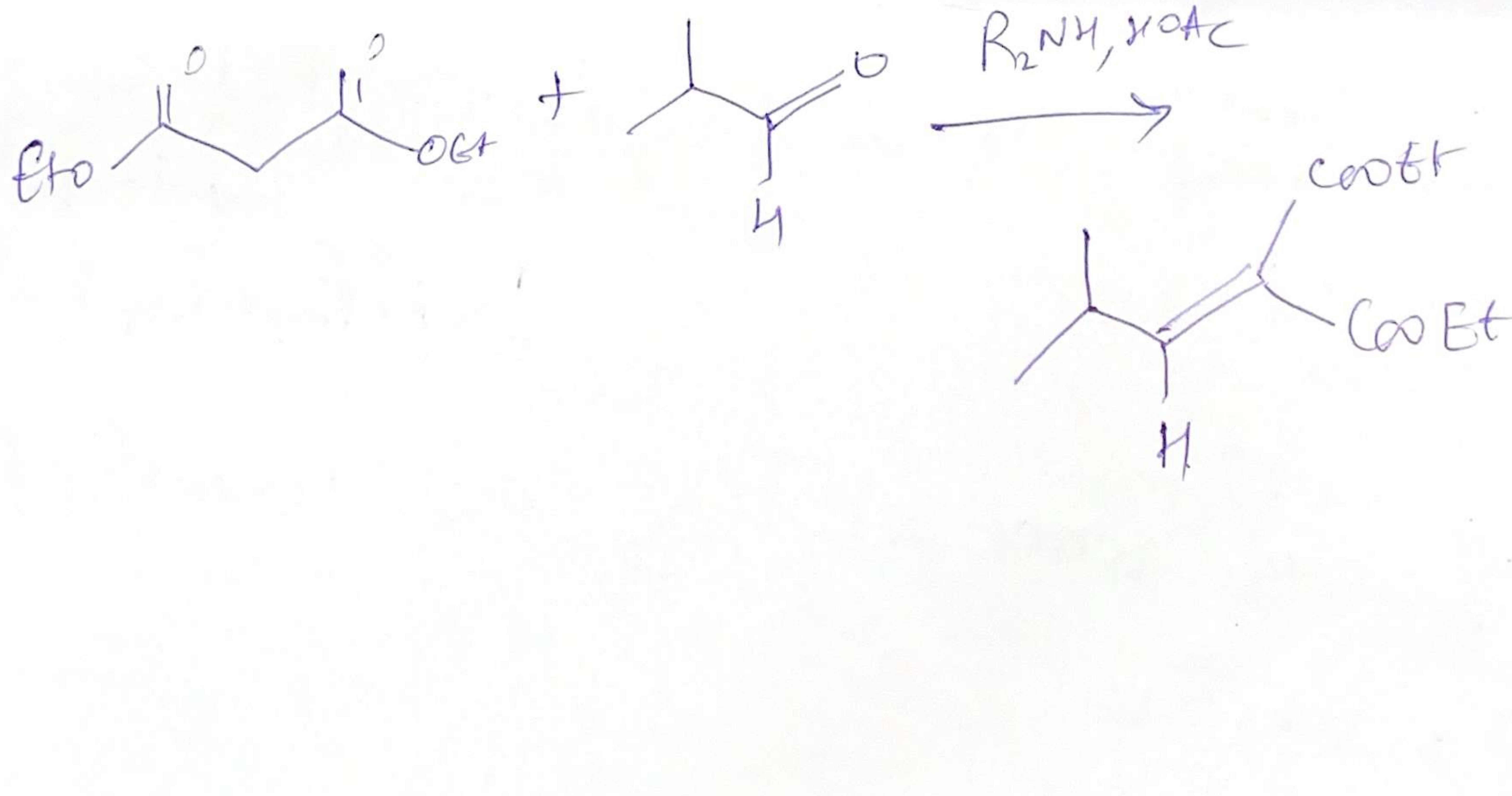
1,3- di Carbonyl ~~Carbo~~ Compounds



Diethyl malonate

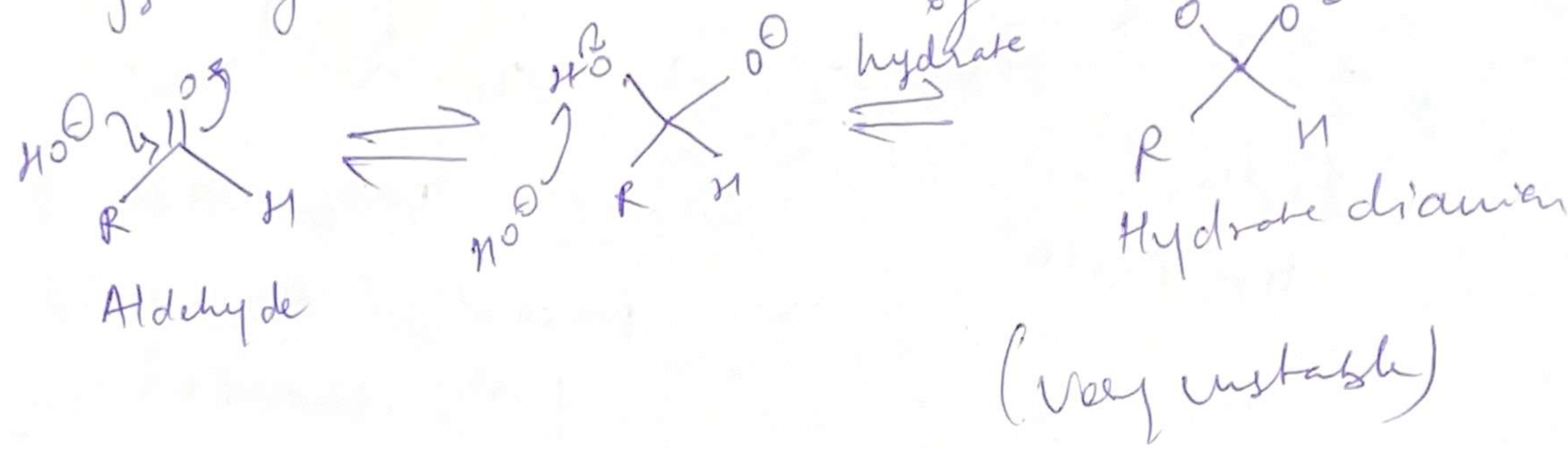
Very stable
Unenolized fraction is poorly electrophilic

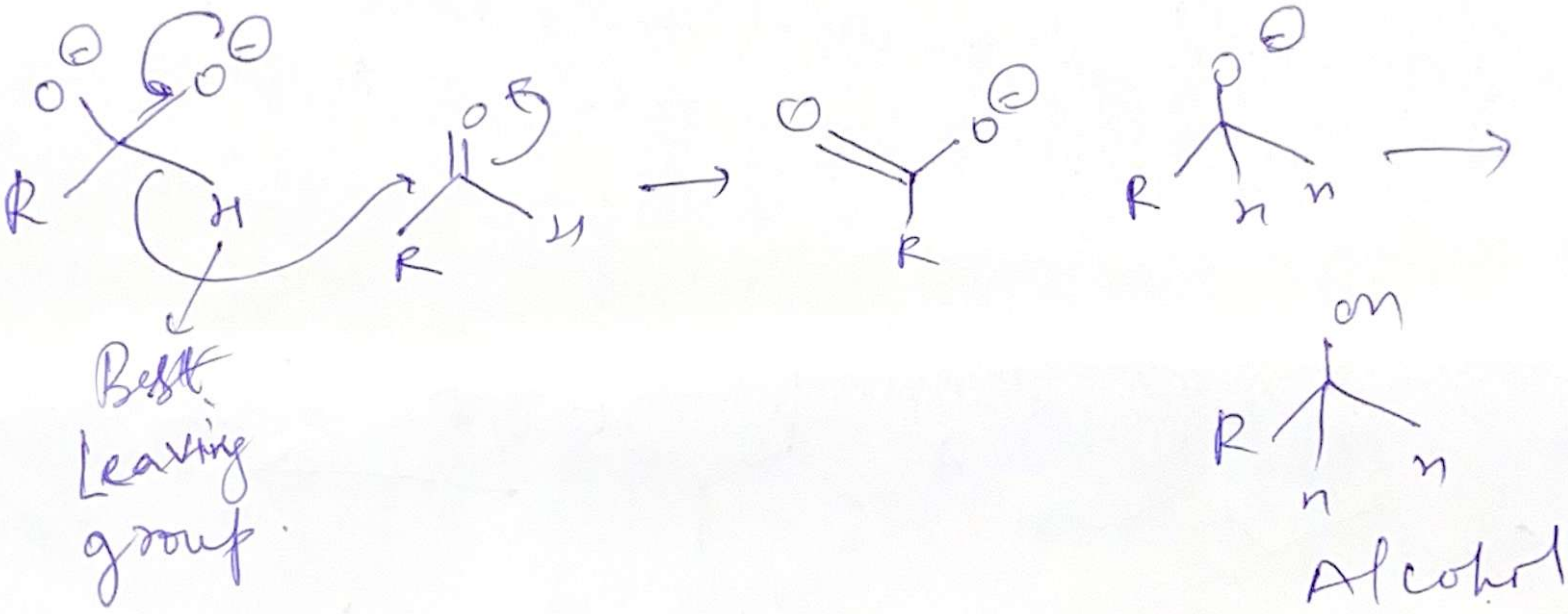
[Large amount of stable enol & a small fraction of unenolized compound]



The Cannizzaro Reaction

Aldehydes are partly hydrated in water which is catalyzed by base.





Disproportionation

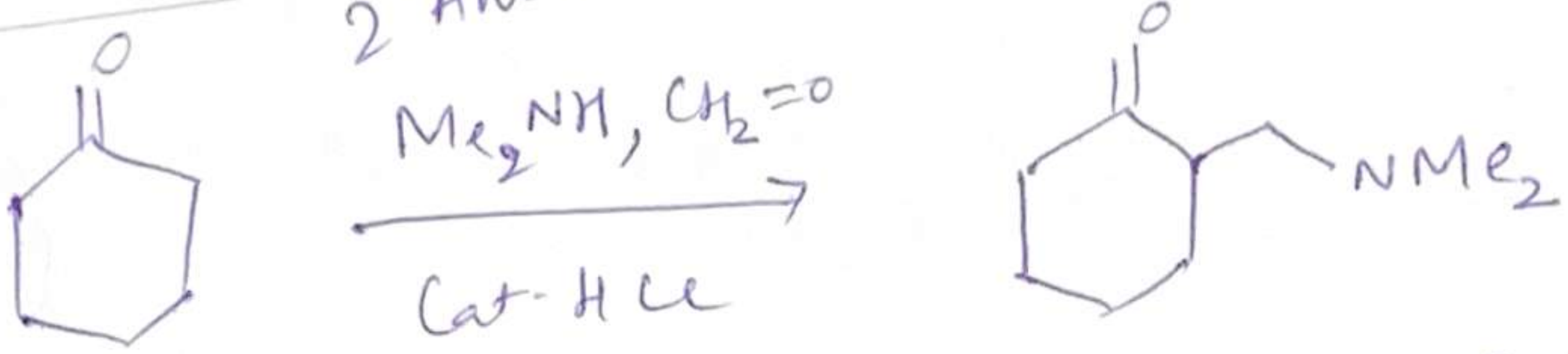
Reaction works only with the unenolizable aldehydes

Mannich Reaction

→ To make amino ketones (drug molecules)

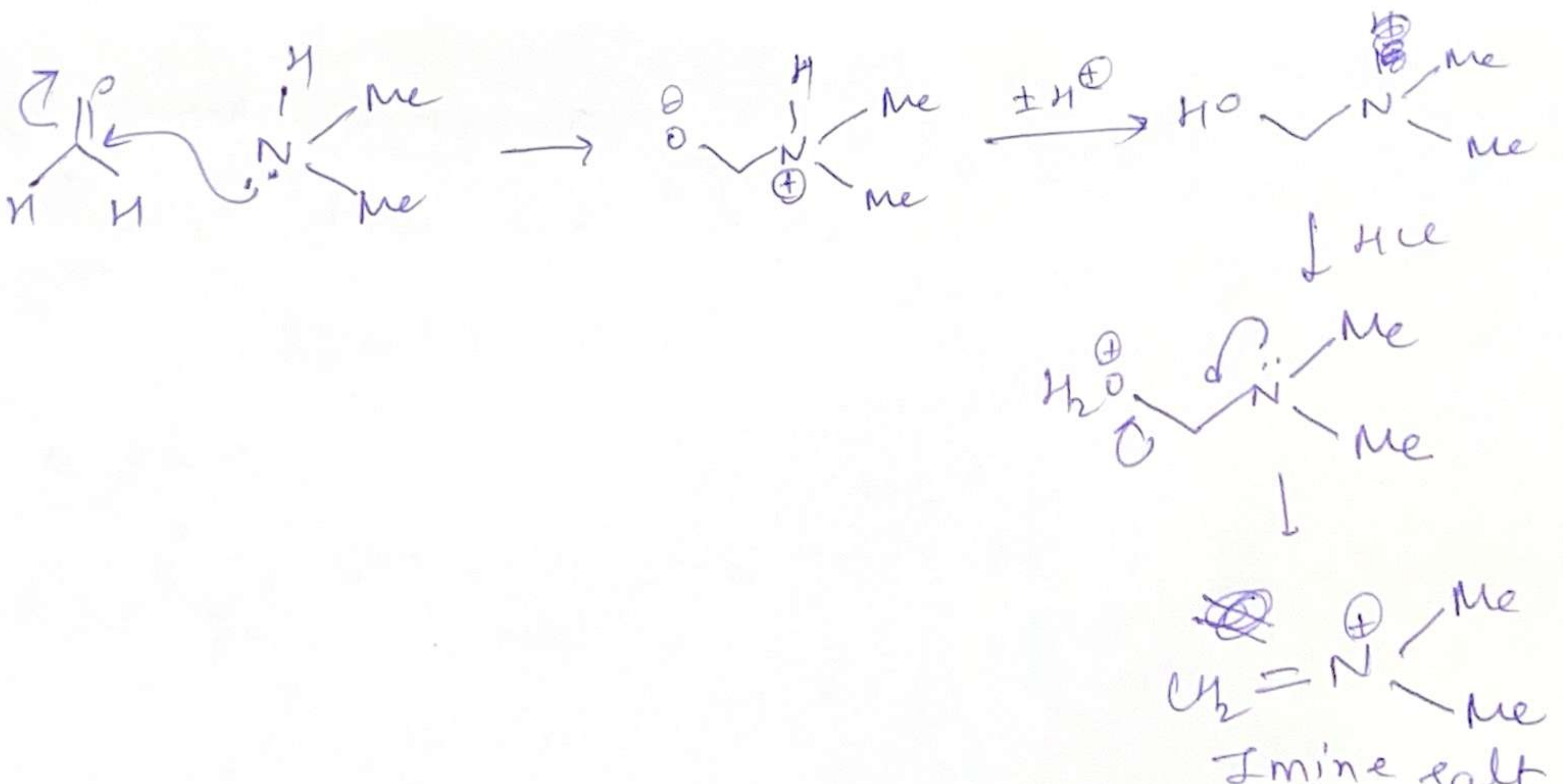
If we use formaldehyde as one of reactant in the aldol, it can lead to Cannizzaro reac^n in the end.

In order to prevent that Mannich reac^n is done.



Enolizable aldehyde or ketone

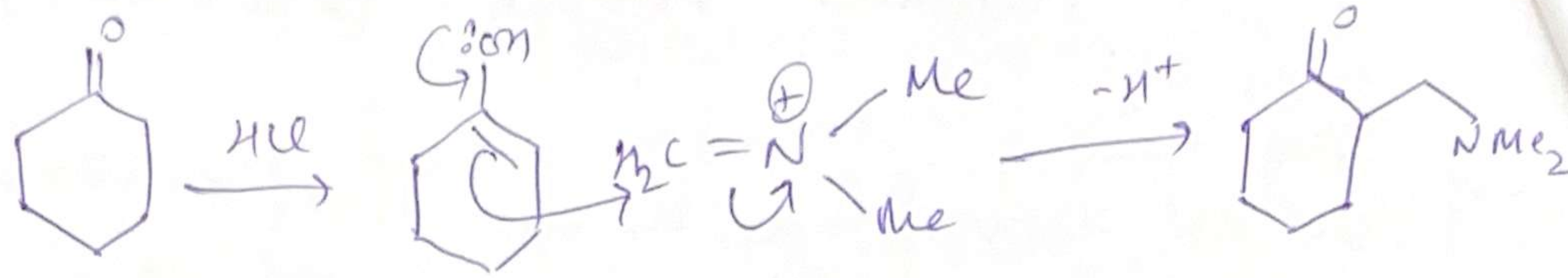
85% yield

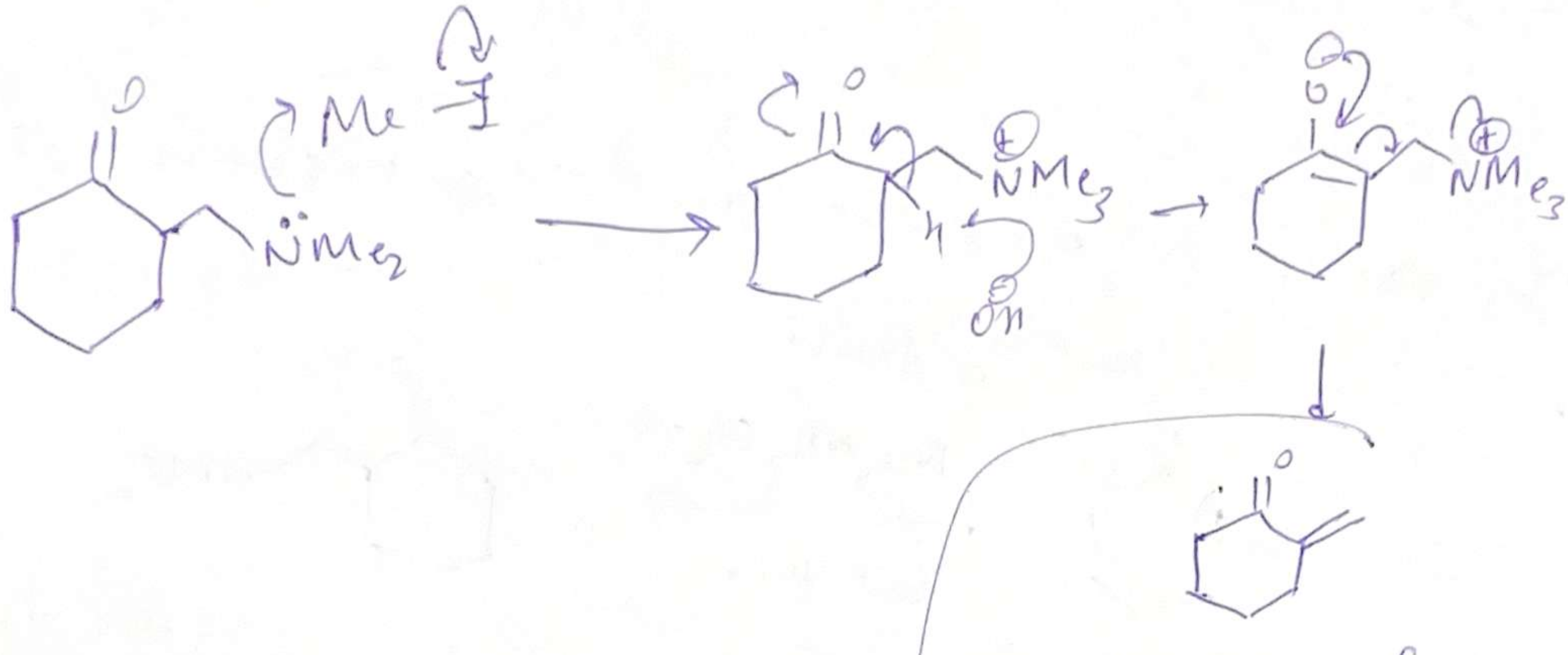


Imine salt

~~Imine~~

Corresponding iodide is sold as Eschenmoser's salt





Exo-methylene compounds

Very reactive

cannot be made by aldol reac^n with formaldehyde

Reformatsky Reaction

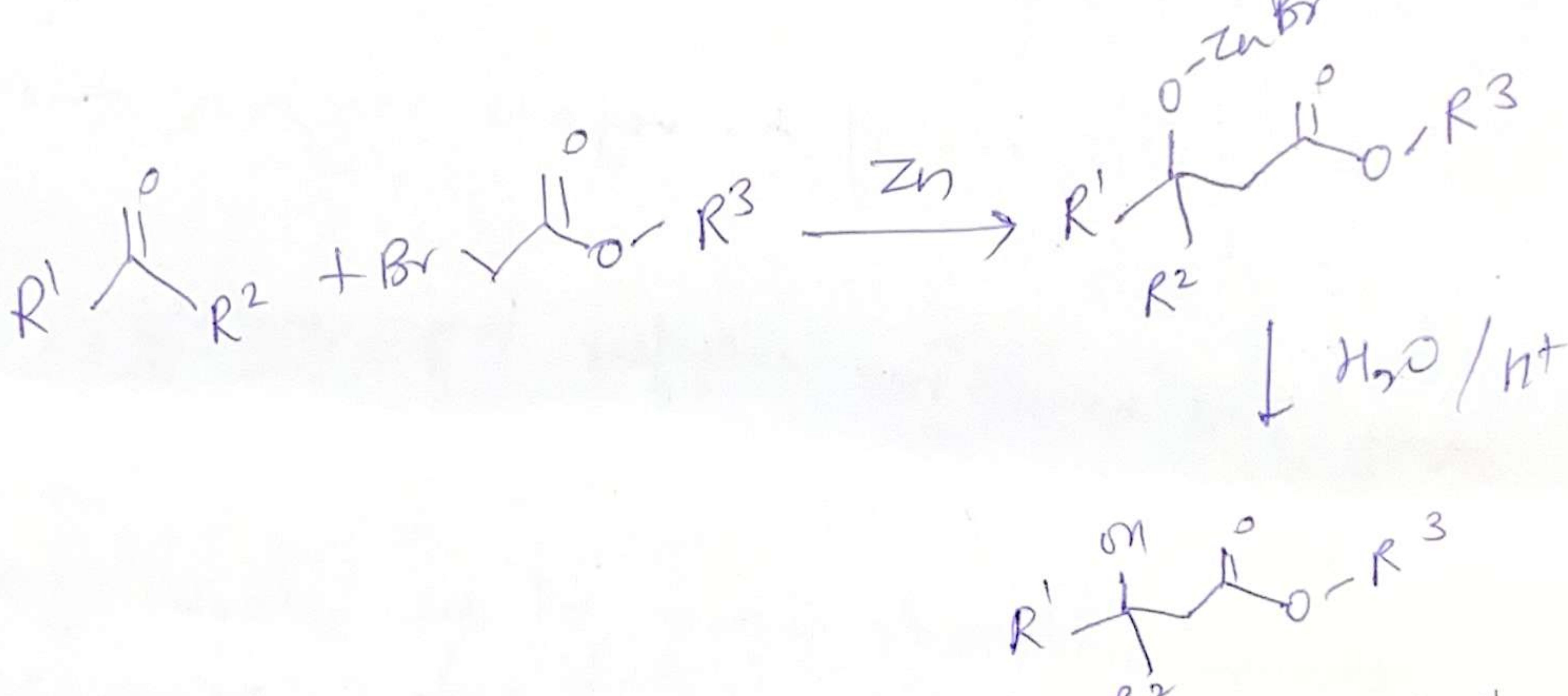
Reacⁿ] Aldehyde or ketones with α-halo esters using metallic Zinc to form β-hydroxy esters

(forms Zinc enolate)

The organozinc reagent is called reformatsky ~~reag~~ reagent/ enolate

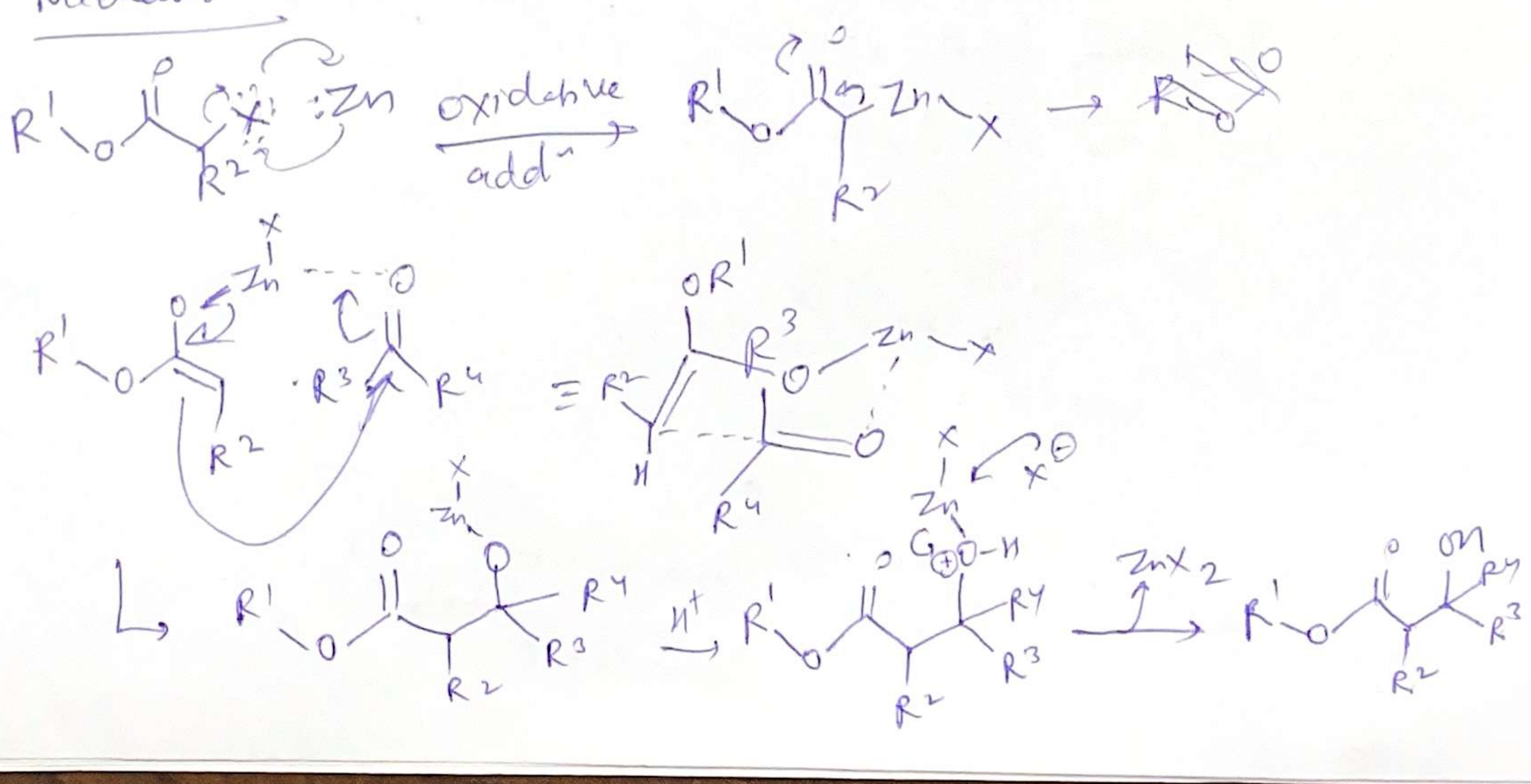
↓

α-halo ester + Zn dust



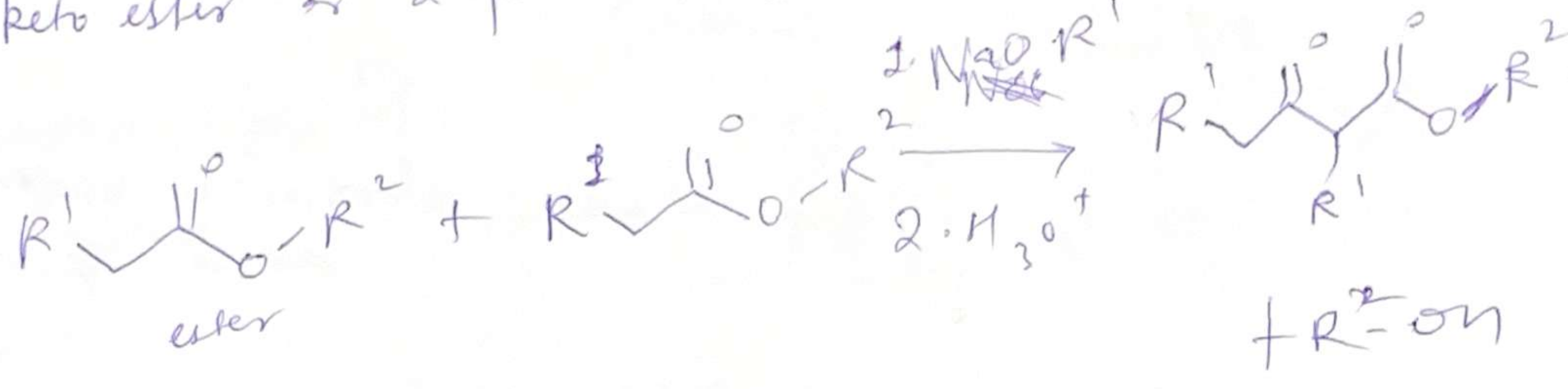
Substrate scope → Acid chloride, imine, ~~nitile~~ nitrile, nitrones

Mechanism



Claisen condensation

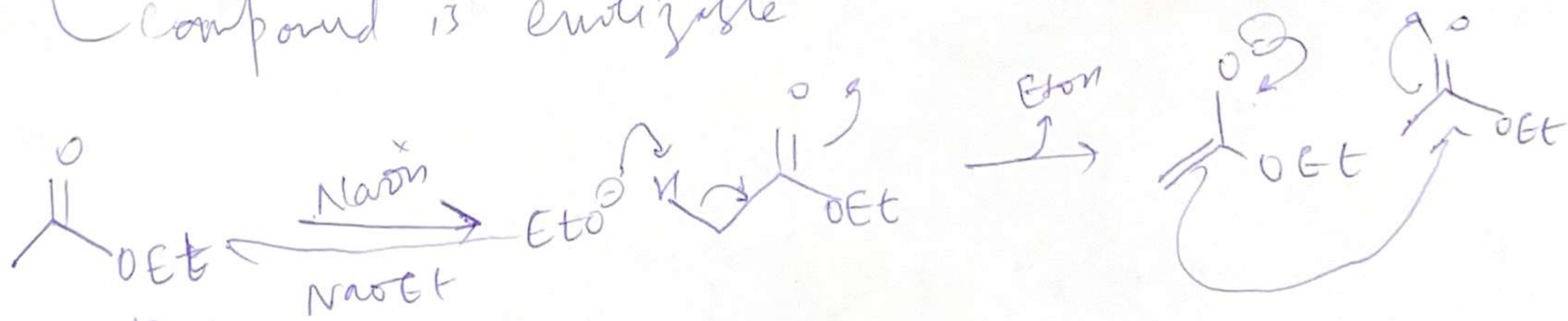
Between two esters or one ester & another carbonyl compound in presence of a strong base to produce a β-keto ester or a β-diketone.



→ At least one of the ~~reagents~~ reagents ~~should~~ must be enolizable.

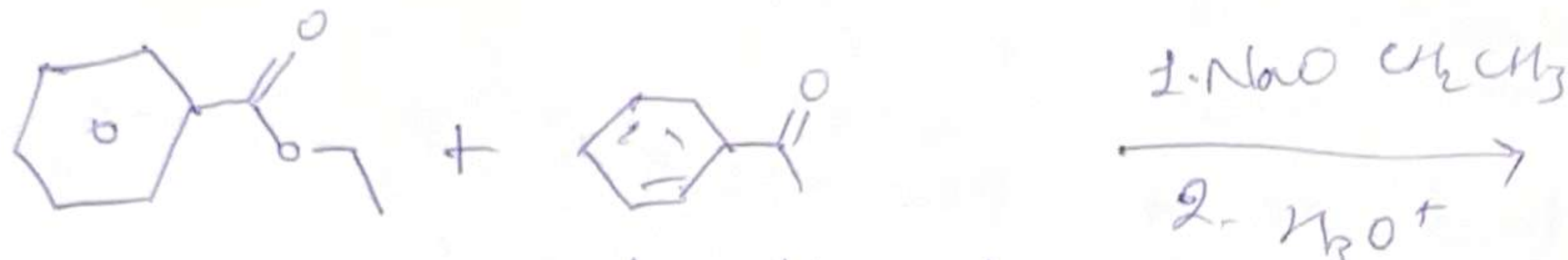
→ Base must not interfere by undergoing Nu substitution or addⁿ. Therefore, the conjugate sodium alkoxide base of the alcohol formed is used, since the alkoxide is regenerated.

In mixed/cross claisen condensation non-nucleophilic base such as LDA since only one compound is enolizable

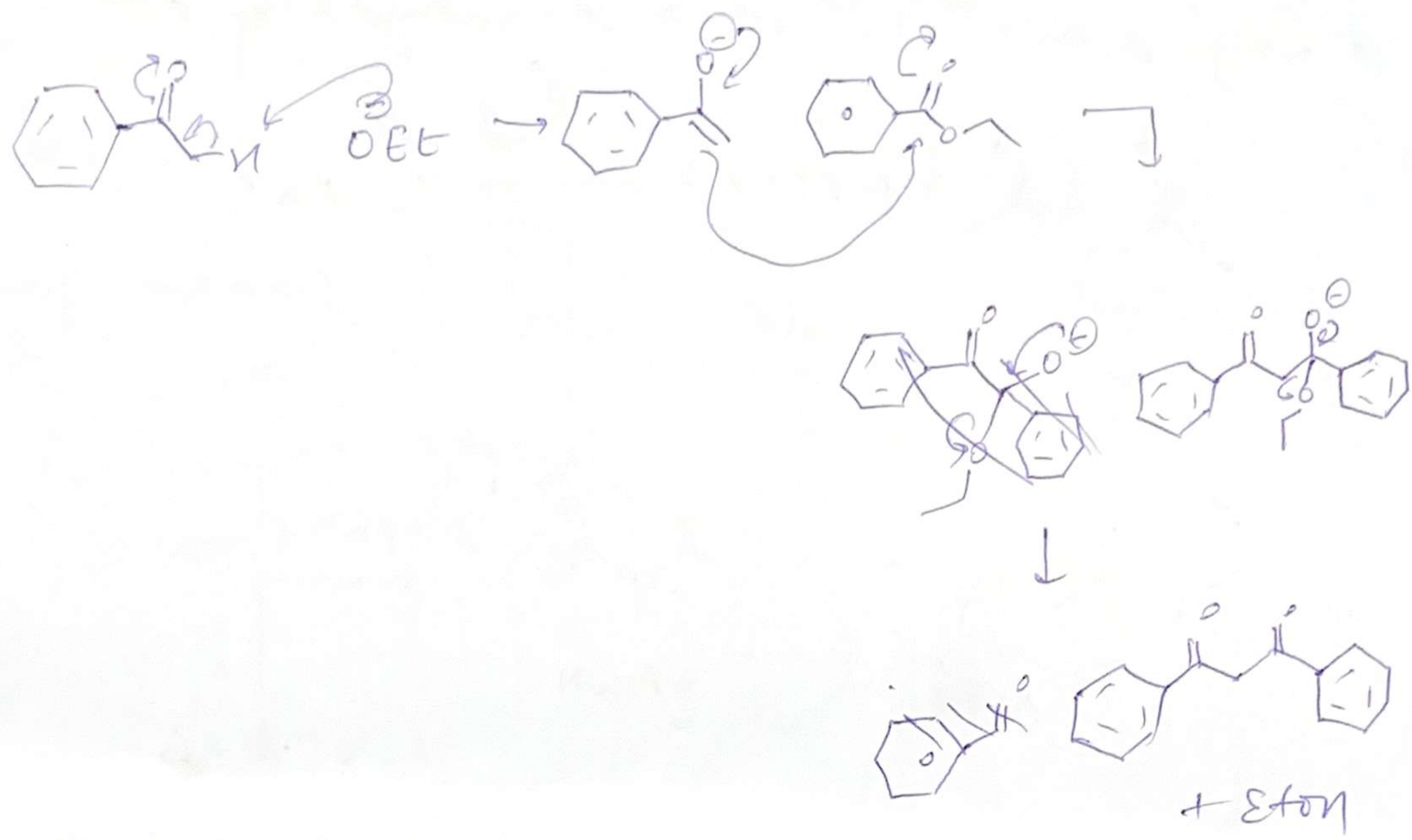


Ethyl acetate

β-keto esters → Acylation at the enolate carbon



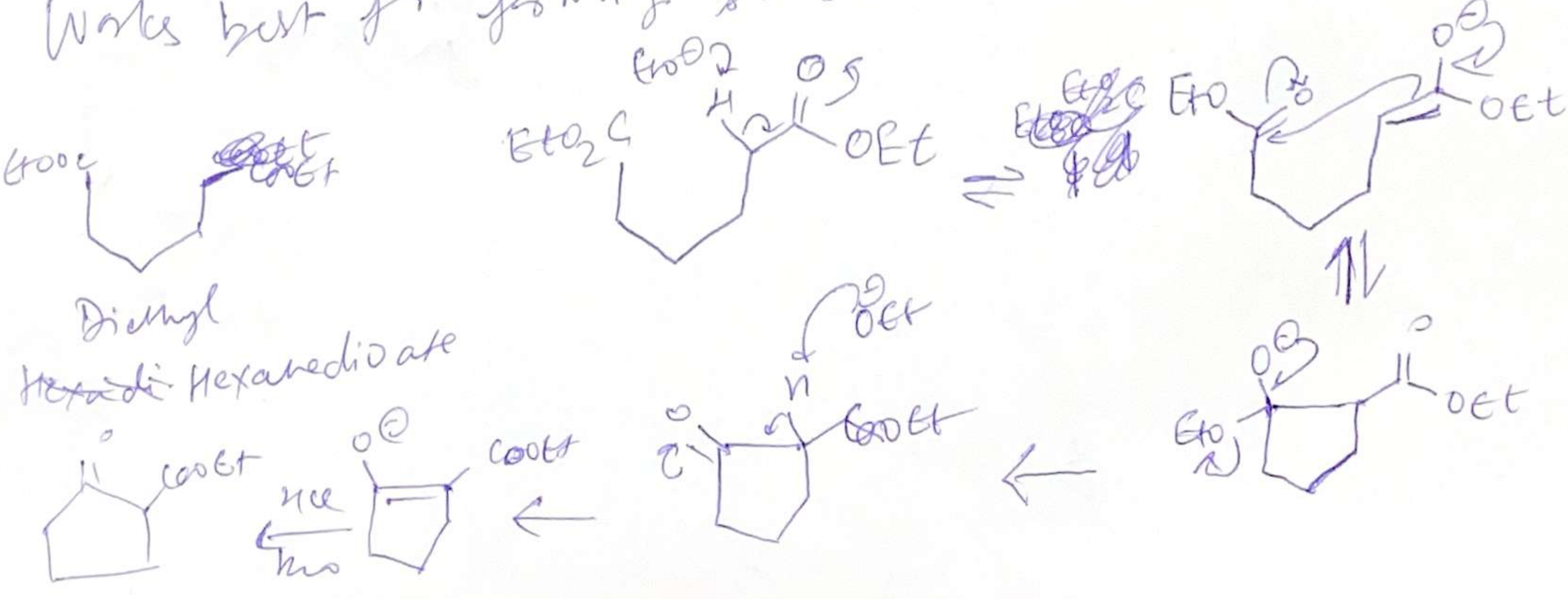
Ethyl Benzoate   Acetophenone



Dieckmann Condensation

Base-Catalyzed intramolecular reacn of a diester that forms a cyclic β-keto ester.

Works best for forming stable 5 & 6-memb. rings.



Diethyl ~~Hexadi~~ Hexanedioate